1 Solution — GIAM 8.2 Problem 1

1.1 Problem

Prove that positive numbers of the form $3k + 1$ are equinumerous with positive numbers of the form $4k + 2$.

1.2 Setup — The Two Sets

Define

$$A \;=\; \{\, 3k+1 \;:\; k = 0, 1, 2, \ldots \,\} \;=\; \{1,\ 4,\ 7,\ 10,\ 13,\ 16,\ \ldots\}$$

and

$$B \;=\; \{\, 4k+2 \;:\; k = 0, 1, 2, \ldots \,\} \;=\; \{2,\ 6,\ 10,\ 14,\ 18,\ 22,\ \ldots\}.$$

Both are infinite arithmetic progressions of positive integers — A starts at 1 with common difference 3, B starts at 2 with common difference 4. The indexing begins at $k = 0$ because $k = 0$ already gives a positive element in each set ($3 \cdot 0 + 1 = 1$ and $4 \cdot 0 + 2 = 2$); any smaller k would produce a non-positive value.

To show $A \equiv B$ — that is, A and B are *equinumerous* — Section 8.1's definition asks us to exhibit a bijection $f : A \to B$.

1.3 The Bijection

Each element of A has a *unique* index k, because $3k_1 + 1 = 3k_2 + 1 \Longleftrightarrow k_1 = k_2$. Hence the rule

$$f : A \;\longrightarrow\; B, \qquad f(3k+1) \;=\; 4k+2 \quad \text{for each } k = 0, 1, 2, \ldots$$

defines a function unambiguously. In words: f sends the k-th element of A (counted from $k = 0$) to the k-th element of B.

1.4 Verifying the Bijection

1.4.1 Injectivity

Suppose $f(3k_1+1)=f(3k_2+1)$. Then $4k_1+2=4k_2+2$, so $4k_1=4k_2$, hence $k_1=k_2$, and therefore $3k_1+1=3k_2+1$. So f is injective.

1.4.2 Surjectivity

Let $m \in B$. By definition of B, there is $k \in \{0,1,2,\ldots\}$ with $m=4k+2$. Set $n=3k+1 \in A$. Then $f(n)=f(3k+1)=4k+2=m$. So f is surjective.

Hence f is a bijection $A \to B$, and so $A \equiv B$. $\blacksquare$

1.5 The Bijection as a Closed Form

Writing f as a formula in n rather than in k: given $n \in A$, we have $k=(n-1)/3$, so

$$f(n) \;=\; 4\cdot\frac{n-1}{3}+2 \;=\; \frac{4n-4+6}{3} \;=\; \frac{4n+2}{3}.$$

The inverse, similarly, is

$$f^{-1}(m) \;=\; \frac{3m-2}{4}.$$

A nice sanity check: the formula $f(n)=(4n+2)/3$ produces an integer *precisely when* $n \in A$. Indeed, if $n=3k+1$, then $4n+2=12k+6=3(4k+2)$, divisible by 3. And $f^{-1}(m)=(3m-2)/4$ produces an integer precisely when $m \in B$: if $m=4k+2$, then $3m-2=12k+4=4(3k+1)$, divisible by 4. So the formulas not only exhibit the bijection on the named sets but also recover the membership conditions for A and B as divisibility constraints.

1.6 Spot Check — First Several Pairings

k	**element of A: $3k+1$**	**element of B: $4k+2$**	**mapped via f**
0	1	2	$f(1)=2$

1	4	6	$f(4) = 6$
2	7	10	$f(7) = 10$
3	10	14	$f(10) = 14$
4	13	18	$f(13) = 18$
5	16	22	$f(16) = 22$
6	19	26	$f(19) = 26$
7	22	30	$f(22) = 30$

Table 1.1

The pairing is the obvious "list-and-match" pairing: index the two sequences in parallel and tie them off by index.

1.7 Alternative Proof — Via $\mathbb{N}$ as Intermediate

Both A and B are equinumerous with $\mathbb{N} = \{0, 1, 2, \ldots\}$ via the same kind of map:

$$g_A : \mathbb{N} \to A, \quad g_A(k) = 3k + 1; \qquad g_B : \mathbb{N} \to B, \quad g_B(k) = 4k + 2.$$

Each g_A, g_B is a bijection (the cancellation-and-construction argument above transfers directly). So $A \equiv \mathbb{N}$ and $B \equiv \mathbb{N}$, and by *transitivity* of $\equiv$ (Section 8.1, Problem 2),

$$A \;\equiv\; \mathbb{N} \;\equiv\; B \quad \Longrightarrow \quad A \;\equiv\; B.$$

In fact, our original $f : A \to B$ is exactly the composite $g_B \circ g_A^{-1}$: given $3k + 1 \in A$, $g_A^{-1}(3k + 1) = k$ and $g_B(k) = 4k + 2$.

So the "direct" bijection and the "via $\mathbb{N}$" bijection are the *same map*; transitivity of $\equiv$ just packages the construction more abstractly.

1.8 Remark — Density vs Cardinality

Inside $\mathbb{N}$ the set A has *natural density* $1/3$ (one out of every three consecutive integers belongs to A), and B has natural density $1/4$. Naively one might think A is "$4/3 \approx 1.33$ times bigger" than B, but this is exactly the kind of intuition that breaks down for infinite sets: cardinality is blind to density. Both sets have cardinality $\aleph_0$, even though one is *thicker* in $\mathbb{N}$ than the other.

The bijection f makes this concrete but does *not* preserve density: f stretches the spacing of elements from 3 to 4. There is no contradiction — the bijection only needs to pair elements off one-for-one, not preserve any ambient structure.

This is exactly the lesson of Section 8.1 made manifest: the equivalence relation $\equiv$ forgets *every* feature of a set except its size, including how the set sits inside a larger ambient set.

1.9 Remark — The General Pattern

The same argument proves a much stronger fact: **any two infinite arithmetic progressions of integers are equinumerous.** For positive common differences a, b and starting values c, d, let

$$A' = \{ak + c \ : \ k = 0, 1, 2, \dots\}, \qquad B' = \{bk + d \ : \ k = 0, 1, 2, \dots\}.$$

Then $f(ak + c) = bk + d$ is a bijection $A' \to B'$ by the proof above, verbatim. The specific case $(a, c, b, d) = (3, 1, 4, 2)$ is Problem 1.

In particular, every infinite arithmetic progression in $\mathbb{N}$ is in bijection with $\mathbb{N}$, hence denumerable.

1.10 Remark — Even More General

The conclusion generalizes further still: *any infinite subset of* $\mathbb{N}$ *is denumerable.* The proof is: well-order the subset (it inherits the well-ordering of $\mathbb{N}$), then list its elements in increasing order $s_0 < s_1 < s_2 < \cdots$; the map $k \mapsto s_k$ is a bijection from $\mathbb{N}$ to the subset.

Arithmetic progressions are simply the cleanest special case — the listing map has a closed-form formula ($k \mapsto ak + c$), so there's no need to invoke the general well-ordering argument.

Viewed this way, Problem 1 is the warm-up for a general principle that becomes powerful in the rest of Chapter 8: most “everyday” infinite sets you can construct out of $\mathbb{N}$ turn out to be denumerable, often via a clever explicit listing. Cantor’s snake (used later in 8.2 for $\mathbb{N} \times \mathbb{N}$) is the dramatic two-dimensional version of the same idea.